

Mitchell Kain

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Portfolio Website: mitchellkain.com

Summary

Recent Mechanical Engineering M.S. graduate specializing in robotics, CAD, and product design, with hands-on experience in SolidWorks, MATLAB, 3D simulation, and robotics. Experience building and managing diverse technical teams and developing aerospace and military systems. Skilled at integrating design, simulation, and control to develop innovative automation and aerospace solutions.

I am an engineer who can calculate, code, model, prototype, and test. I bring curiosity, grit, and results.

Skills

Coding: MATLAB, Python, SQL, Linux, C/C++, Markdown, HTML, LaTeX, G-code

Software: ROS 2, Arduino, ESP32, 3D-printing slicers

- **Computer-Aided Design:** SolidWorks, OnShape, FreeCAD, MeshMixer, Blender
- **Simulation and Analysis:** Abaqus, COMSOL, Simulink

Fabrication: 3D Printing, Rapid Prototyping, Molding, Soldering, Laser Cutting, CNC, Hand/Power Tools

Embedded & Sensing: ESP32, I2C/UART, Hall-effect sensors, Magnetic sensing, IMU, LiDAR

Other: Leadership, Innovation, Problem Solving/Design, Research, Technical Writing, Chess

Education

Boston University, MS in Mechanical Engineering, Dynamic Systems and Controls

Jan 2025

- **Coursework:** Robot Motion Planning, Vibrations of Complex Mechanical Systems, Soft Robotics, Additive Manufacturing, Fluid Dynamics, Thermodynamics, Static Analysis, Dynamic Analysis
- **Lab work:** Developing and rapidly prototyping soft robots in BU's Morphable Biorobotics Lab (MBL), and Robotics & Autonomous Systems Teaching and Innovation Center (RASTIC)
- **Graduate Course GPA:** 3.92 (3.84 all classes)

Galvanize Inc., Data Science Boot Camp – New York, NY

Mar 2020 – Jun 2020

USMC Primary Flight School, Second Lieutenant: Student Naval Aviator – Pensacola, FL

Oct 2015 – Jan 2017

- Intensive Military Leadership and Aviation training, including: Navigation, Radio communication, Hydraulic Systems, Engine Systems, Aerodynamics, and Military Operations
- Live flight training hours in Cessna 172 and T6-Texan acrobatic aircraft and simulators

University of Rochester, BS in Applied Mathematics

May 2011

Experience

NASA Langley Research Center, Aerospace Academy Lead Research Associate – Hampton, VA

Jun 2024 – Aug 2024

- Led a diverse team of 12 Mechanical, Electrical, and Aerospace Engineers and Physicists to research new technologies and build prototypes, enhancing the capabilities of the NASA Aerospace Research Directorate
- Designed and tested Emergency Drone Recovery and Detection Systems, improving safety measures
- Used MATLAB and video analysis to analyze descent profiles and impact data to optimize impact reduction

Galvanize Inc., Lead Data Science Resident / Curriculum Developer – New York, NY

Jun 2020 – Aug 2021

- Revised curriculum, tutorials, and lectures on Python, SQL, Statistics, Probability, Machine Learning, Computer Vision, Data Engineering, ensuring the most up-to-date and accessible material was delivered
- Taught Data Science and Python to over 40 Ph.D., Master's, Professional, and Veteran students equipping them with essential skills for careers in data science
- Specialized in Computer Vision applications and Reinforcement Learning, contributing to innovative projects and enhancing the curriculum with cutting-edge techniques

United States Marine Corps, Captain: Supply and Logistics Officer – Camp Lejeune, NC

Jan 2017 – Jan 2020

- Supervised all ground supply operations for the 2nd Marine Aircraft Wing (2nd MAW), ensuring financial obligations were met and unit readiness remained high
- Mentored and managed 16 officers while leading inspection teams to ensure compliance with all regulations, standing orders, and federal law

- Oversaw \$77 million in military equipment and a \$2 million budget, maintaining mission capability and accountability across multiple commands
- Procured, implemented, and integrated new technologies as Innovation Officer, improving operational efficiency and informing future Marine Corps procedures

Projects

<u>Chess Robot / Plotter / Laser Engraver</u>	Sep 2025 – Current
• Designing and building a custom XY gantry robot for plotting, laser engraving, and autonomous chess play • Motion system driven by three stepper motors using Arduino Uno running GRBL, coordinated by Raspberry Pi 5 (Ubuntu) • Developing interchangeable end effectors and board-state sensing for human–robot chess interaction, with planned ROS 2 integration • Evaluating and prototyping board-state sensing approaches (magnetic, Hall-effect, and RFID) with emphasis on robustness and manufacturability	
<u>NASA Guided Parachute</u>	Jun 2024 – Aug 2024
• Designed, built, and flight-tested a steerable gliding parachute to reduce the impact energy of a large falling drone for the NASA UAS Office • Enabled controlled descent and lateral guidance to avoid populated areas, infrastructure, and environmental hazards	
<u>Soft Robotics Competition</u>	Mar 2024 – May 2024
• Designed and fabricated a primarily silicone, pneumatically actuated soft robot that placed 1st in an intercollegiate competition • Six-legged robot successfully traversed an obstacle course including turns, ramps, water, sand, and drop tests	
<u>LiDAR-Based Autonomous Hovercraft</u>	Oct 2023 – Dec 2023
• Built a three-fan robotic hovercraft using LiDAR and inertial sensing for autonomous navigation • Implemented onboard control on a Raspberry Pi to compensate for drift using potential-field-based path planning • Demonstrated applicability to uneven or amphibious environments such as security, mapping, and search-and-rescue	